PRECALCULUS · UNIT 3: CONICS

Science Museum Conics

Design a science museum exhibit with a parabolic dish, an elliptical whispering gallery, and a hyperbolic tower.

Honors

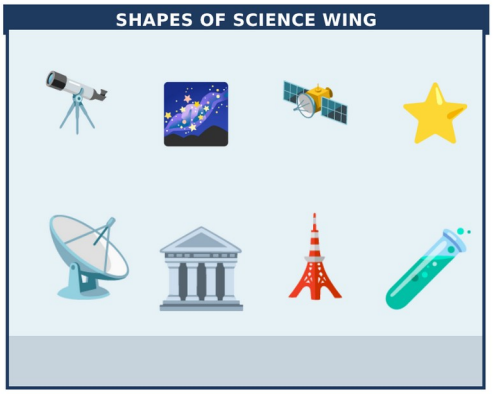
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

The city science museum is adding a "Shapes of Science" wing. Each exhibit is built around a conic section: a round planetarium, a parabolic dish, an elliptical whispering room, and a hyperbolic cooling tower model.

You will write the equations, locate the foci, and make the measurements the builders need.



Dish
Diameter 12 ft
Depth 1.5 ft

Gallery
Ellipse 60 ft long
40 ft wide

Learning Goals

- I can derive and use conic equations from geometric definitions.
- I can solve design problems involving foci, directrices, and eccentricity.
- I can solve systems involving conics.

Project Timeline

When	What you will do
Day 1	Parabolic Engineering
Day 2	Gallery & Planetarium
Day 3	Hyperbolic Navigation
Final	Exhibit Design Packet + reflection

Part 1: Parabolic Engineering

Day 1 · Use the Specification Cards.

Exhibit Specification Cards

Planetarium	Satellite Dish	Flashlight
Circle, center (0, 0)	Diameter 12 ft	Parabolic mirror
radius 25 ft	depth 1.5 ft	$y = \frac{1}{8}x^2$

Whispering Gallery	Cooling Tower	Navigation
Ellipse 60 ft by 40 ft	Hyperbola, narrowest	Two stations
	width 30 ft	100 mi apart

1. Use the definition of a parabola (equal distance from focus (0, p) and directrix y = −p) to derive $x^2 = 4py$.

2. A dish has focal length 4 ft and must be 1.25 ft deep. What diameter should it have?

3. A parabolic arch over the entrance is 30 ft wide and 18 ft tall. Write its equation with the vertex at the top (0, 18).

4. Can a 12-ft-tall truck that is 10 ft wide pass under the arch? Show your work.

Part 2: Gallery & Planetarium

Day 2 · Ellipse gallery 60 ft by 40 ft; planetarium circle radius 25 ft centered at (0, 0).

5. The museum wants the gallery foci exactly 40 ft apart and the room 50 ft long. Find the width.

6. Find the points where the circle $x^2 + y^2 = 625$ and the ellipse $\frac{x^2}{900} + \frac{y^2}{400} = 1$ intersect.

7. What does that intersection mean if the planetarium and gallery overlap on a floor plan?

8. Find the eccentricity of the new gallery from the first question and compare it to the original.

Part 3: Hyperbolic Navigation

Day 3 · Stations A(−50, 0) and B(50, 0), units in miles.

9. A ship receives B's signal 0.000322 s before A's. Signals travel 186,000 mi/s. Find the difference in distances and the hyperbola.

10. The ship is also on the line y = 20. Find its position (it is closer to B).

11. Write a design memo explaining one conic exhibit, with equations and measurements.

Final Product: Exhibit Design Packet

Your final product must include:

- Parabola derivation and arch clearance
- Ellipse redesign and circle intersection
- Hyperbolic navigation fix
- Eccentricity comparison
- Design memo

Reflection

R1. How does each conic's geometric definition lead to its useful property?

R2. Why does eccentricity describe the shape of an ellipse?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Conic equations, foci, and measurements are correct and used to design working exhibits. Goes beyond the minimum with a creative, realistic choice.	Conic equations, foci, and measurements are correct and used to design working exhibits.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Exhibit packet includes equations, labeled sketches, and builder measurements. Polished, organized, and visually engaging.	Exhibit packet includes equations, labeled sketches, and builder measurements. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.